

教學大綱 Syllabus

科目名稱：商業資料分析與機器學習

Course Name:
Data Science and Machine Learning for Business

學年學期：115-1
Fall Semester, 2026

科目代碼：301767001
Course No.301767001

課程簡介

Course Description



修別：
選

Type of
Credit:
Elective

3.0

學分數
Credit(s)

30

預收人
數

Number
of
Students

This course introduces practical data science and basic machine learning for business students with little technical background.

The first part develops core data skills such as working with data tables, cleaning data, visualization, and reproducible workflows, following the practical approach of ***Coding for Economists*** book. The second part introduces simple machine learning concepts and business applications in marketing, finance, and operations. The focus is on using data to support decision-making rather than on advanced mathematics or programming.

The class is for undergraduate students with programming skills, basic mathematics, and are curious to learn about how machine learning can be applied for business decision-making.

課程資料

Course Details

開課單位：國貿三甲、國貿三乙、國貿四甲、
國貿四乙

Course Department: International Business/B/3&4

授課老師：裴典富

Instructor: BUI DIEN GIAU

先修科目：無

Prerequisite(N/A)

上課時間：二234

Session: tue09-12

核心能力分析圖

Core Competence Analysis Chart

能力項目說明

Students will be able to:

課程目標與學習 成效

- Understand key data science and machine learning concepts for business.
- Organize, clean, summarize, and visualize business data.

- Use basic version control and database tools for managing business data projects.
- Apply simple models to business problems.
- Interpret model outputs for decision-making.
- Recognize the limitations and risks of machine learning in practice.

課程屬性

Course Characteristics

級分說明 Score Description:

1關聯性最弱，2關聯性稍弱，3關聯性中等，4關聯性稍強，5關聯性最強。(1-5非百分比，而是Rank)
1 indicates weakest relevance, 2 indicates slightly weaker relevance, 3 indicates moderate relevance, 4 indicates slightly stronger relevance, and 5 indicates strongest relevance. (1-5 does not represent a certain percentage but indicates the rank).

課程屬性項目 Course Characteristics Items	級分 (1~5) Ranking	說明 Description
人文關懷課程 Humanity and Care Course	1	課程教學內容涉及學生人文關懷素養與態度之培養。 The course contents include the development of students' abilities and appreciation for humanistic care.
跨領域課程 Interdisciplinary Course	5	課程教學包含不同學門、領域之內容。 The courses include contents from different disciplines and fields.
社會參與課程 Social Engagement Course	1	課程教學內容涉及社會參與、服務等。 The course contents include social engagement and services.
總整課程 Capstone Course	3	係系統整與深化學生歷年所學，並運用於實務操作與討論，藉以檢視和評估學生的整體學習成效。 The courses are designed to integrate and intensify students' academic results from previous years and apply them in practice and discussions for the purpose of reviewing and evaluating the overall effectiveness of students' learning.
永續課程 SDGs	3	課程主軸明確探討永續性之統整概念及/或聯合國永續發展目標及/或「生態」與「社會/文化/經濟」系統之間的相互依賴性。關聯度級分請點選連結說明。 A unit of instruction that has a primary and explicit focus on the integrated concept of sustainability, and/or UN Sustainable Development Goals, and/or the interdependence of ecological and social/cultural/economic systems. Please refer to the link of relevance score.



課程屬性項目 Course Characteristics Items	級分 (1~5) Ranking	說明 Description
資訊判讀課程 Information Literacy Course	4	課程內容旨在教授媒體、網路相關專業資訊之真偽辨別與判讀能力。 The course would provide skills to evaluate the quality of information found on the media and the Internet.
AI 素養課程 AI Competence Course	5	將 AI 知能運用融入課程，引導師生與 AI 共學、協作，多元培養學生 AI 素養。 Integrate the application of AI knowledge into the curriculum, lead teachers and students to learn and collaborate with AI, and cultivate students' AI literacy.

每周課程進度與作業要求

Course Schedule & Requirements

Week	Topic	Content and Reading Assignment	Teaching Activities and Homework
1	Introduction to Data Science for Business: data, decisions, and business questions		
2	Coding Basics and Workflow: notebooks, scripts, and reproducible work		
3	Git, GitHub, Version Control, and Databases (SQL) for Business Projects		
4	Working with Data Tables: importing, filtering, sorting, and summarizing data		
5	Data Cleaning and Transformation: missing values, reshaping, and creating variables		



Week	Topic	Content and Reading Assignment	Teaching Activities and Homework
6	Data Visualization for Business: charts, dashboards, and clear communication		
7	Combining and Collecting Data: merges, simple data gathering, and project workflow		
8	Midterm exam		
9	What Is Machine Learning in Business		
10	Introduction to Prediction (Regression)		
11	Model Evaluation (Accuracy and Overfitting)		
12	Classification		
13	Extension: Trees and neural networks; Ethics, Bias, and AI in Business		
14	Applications in Marketing (Targeting and Recommendation)		
15	Applications in Finance and Operations (Credit Risk, Fraud, Demand Forecasting)		
16	Final Project Presentations		

授課方式

Teaching Approach

30%

講述
Lecture

30%

討論
Discussion

40%

小組
活動
Group
activity

0%

數位
學習
E-
learning

0%

其
他：
Others:



評量工具與策略、評分標準成效

Evaluation Criteria

Midterm exam: 30%

Problem sets: 30%

Final Project: 40%

Final Project: Students will analyze a simple dataset and apply a basic model to solve a business problem, such as customer segmentation, sales prediction, or demand forecasting.

Deliverables: short report and presentation.

指定/參考書目

Textbook & References

1. Turrell, Arthur. Coding for Economists (selected practical chapters) **[C4E]**
2. James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani, eds. 2013. An introduction to statistical learning: with applications in R. Springer texts in statistics 103. New York: Springer. **[ISLR]**
3. Instructor-provided datasets and case studies

已申請之圖書館指定參考書目

本課程可否使用生成式AI工具

Course Policies on the Use of

Generative AI Tools

有條件開放使用: it is allowed in-class with teacher permission. No AI usage is allowed during the exams.
Conditional Permitted to Use

課程相關連結

Course Related Links



課程附件

Course Attachments



課程進行中，使用智慧型手機、平板等隨身設備

To Use Smart Devices During the
Class

其他:Only laptop or tablet (for notetaking) are allowed. No smartphones in class.
Other regulation

授課教師Office Hours及地點

Office Hours & Office Location

Wednesday, 12:00-16:00 PM, Office Room: TBA

教學助理基本資料

Teaching Assistant Information

